

3D reconstruction and characterization of cotton bolls in situ based on UAV technology

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ABSTRACT

Phenotypic traits at the organ scale hold significant importance in the realm of plant breeding, notably in evaluating genetic diversity, selecting innovative cultivars, and forecasting potential yield. Ground platforms, despite delivering accurate results in organ-scale phenotypic studies, frequently exhibit limitations concerning efficiency and adaptability, and can adversely affect the soil. The Unmanned Aerial Vehicle (UAV) platform emerges as an innovative medium, distinguished by its high-throughput and adaptability, thus offering immense prospects in phenotypic research. Nonetheless, its precision within the organ scale warrants further investigation. As a notable economic plant species, cotton (*Gossypium* spp.) provides the textile industry with its essential raw material through the fruit it bears, the cotton boll. This study, employing the characterization of cotton boll breeding materials as an example, scrutinizes the capability of UAV platforms in executing precise enumeration and yield estimation of three-dimensional (3D) cotton bolls in agricultural fields. An innovative route, the Cross-circling oblique (CCO) route, was employed in conjunction with lightweight UAVs for reconstructing a 3D point cloud of cotton bolls in the field, and juxtaposed against the traditional nadir route. A comprehensive process was also developed for characterizing the quantity, volume, and spatial distribution of cotton bolls, further facilitating yield estimation. The CCO-produced point cloud demonstrated a superior completeness compared to the nadir-produced point cloud, proving capable of effectively mitigating occlusion in the middle and lower sections, thereby procuring a more comprehensive 3D cotton boll. In terms of cotton boll counting ($R^2 = 0.92$ vs $R^2 = 0.73$) and yield estimation (seed cotton: $R^2 = 0.7$ vs $R^2 = 0.62$, lint cotton: $R^2 = 0.75$ vs $R^2 = 0.55$), the precision of the CCO-derived point cloud surpassed that of the nadir-derived point cloud considerably. Notably, yield estimations were more precise when based on cotton boll count for seed cotton, and on boll volume for lint cotton. Utilizing the high-precision CCO-produced cotton boll point cloud, an analysis was conducted on the spatial distribution of cotton bolls for all breeding materials, and the yield was subsequently projected. The contribution of this study lies in the high-throughput capture of organ-scale traits of field cotton bolls using the UAV-based CCO route, enabling accurate yield estimation. This approach may also be applicable to extracting organ-scale traits of other field crops. It demonstrates the immense potential of combining UAV platforms with the CCO route for high-throughput acquisition of crop organ-scale traits, which could become a powerful tool in plant breeding.

1. Introduction

Phenotypic traits at the organ scale play a crucial role in plant breeding (Jin et al., 2021; Song et al., 2021). These traits, which encompass characteristics such as the size, shape, and function of individual plant organs including leaves, flowers, and fruits, serve as crucial

indicators of a plant's health, productivity, and adaptability (Marsh et al., 2021). Such invaluable data can guide breeders in identifying and preserving genetic variations, assist in the selection of plants with favorable traits, and offer substantial utility in forecasting potential yields (Maman et al., 2004; Boyles et al., 2017).

Cotton (*Gossypium* spp.), a paramount fiber crop, yields high-quality

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fibers integral to various cotton textile production (Ye et al., 2023). Concurrently, its seeds are substantial sources for oil extraction and feed production, underlining cotton's vital role across fiber, oil, and feed industries (Wang et al., 2023). Cotton bolls, the fruit of cotton, serve as crucial metrics for assessing cotton yield and as direct indicators for evaluating qualities such as cotton's quality, resistance to diseases and pests, among other significant traits. Such traits inherently impact cotton's economic value and cultivation benefits. The cotton bolls phenotyping decisively influences cotton's genetic enhancement, novel variety screening, and production management. A thorough comprehension of the cotton boll phenotyping enables more accurate prediction and evaluation of cotton yield and quality, facilitating the selection of superior cotton varieties and underpinning the science of cotton cultivation management (Conaty et al., 2022). A major challenge is that prevailing cotton boll phenotyping are predominantly reliant on manual measurement and observation (Sun et al., 2020). This methodology, while laborious and time-consuming, often fails to meet the demands of modern cotton production for accuracy and efficiency (Jung et al., 2018). An exigent necessity is the development of a high-throughput cotton boll traits analysis method to rapidly obtain abundant, accurate cotton boll traits. This progress would bolster the enhancement of cotton variety selection and management strategies.

Crucial characteristics of cotton encompass the number, volume, distribution, and yield of cotton bolls (Sun et al., 2020). Predominantly, two-dimensional (2D) image-based technologies have been adopted for identifying cotton bolls, such as a field detection method found on regional semantic image segmentation proposed by Li et al. (2016) and weakly supervised deep learning models to segment and count cotton bolls from proximal imagery proposed by Adke et al. (2022). Nevertheless, these techniques only capture the number of cotton bolls within an image's range, and fail to provide the number at the plot scale or reveal other traits. This 2D image technology extends to recognizing other plant organs, such as in the detection of mango fruit (Stein et al., 2016) and apples (Bargoti and Underwood, 2017). However, the efficacy of such image-based methods is inherently limited by their 2D nature, which precludes them from addressing issues in three-dimensional (3D) space, like the occlusion problem among plant organs (Xiao et al., 2020; Xiao et al., 2021). Consequently, when two cotton bolls are closely adjacent or one is obscured by another, 2D image technology may falter in the correct identification and counting process. This limitation inevitably constrains the accuracy and reliability of these technologies in estimating and analyzing cotton boll characteristics.

Sensing technologies that use 3D are primarily mounted on ground mobile platforms to obtain high-precision 3D structures of plant organs at close range (Deery et al., 2020). Light detection and ranging (LiDAR) are the most commonly used 3D sensor for characterizing plant organ scales in field conditions (Yan et al., 2020; Zhou et al., 2020). For example, Malambo et al. (2019) placed a terrestrial LiDAR on a sprayer to capture point clouds of sorghum panicle in the field and accurately estimate their characteristics (panicle length: Pearson correlation (r) = 0.88, root mean square error (RMSE) = 3.10 cm; panicle width: r = 0.79, RMSE = 1.67 cm; plant height: r = 1.00, RMSE = 0.80 cm). LiDAR offer high precision but are expensive. Another type of 3D sensor is the depth camera. For example, Dube et al. (2020) installed multiple RealSense depth cameras on a field vehicle to collect 3D data of cotton bolls during three seasons. Based on the distribution and cumulative frequency of boll structures, the irrigation volume and the impact of varieties on the distribution of cotton bolls were determined. Depth cameras are relatively cheaper than LiDAR but have lower resolution and are susceptible to sunlight interference (Kazmi et al., 2014). Multiple-view geometry methods are passive 3D reconstruction techniques that can reconstruct 3D point clouds from RGB images with overlapping views (Hui et al., 2018; Gil-Docampo et al., 2020). They offer low cost and high precision. Researchers often mount multiple RGB sensors on field mobile platforms to acquire a collection of plant multi-view images and reconstruct highly accurate 3D plant structures using multiple-view geometry methods

(Salas Fernandez et al., 2017; Sun et al., 2020). For example, Sun et al. (2020) mounted five RGB cameras respectively on the top and sides of a field vehicle to capture multi-view images of cotton plants and reconstructed a 3D point cloud of the cotton plants. Additionally, they used clustering-based methods to analyze the number, distribution, and yield of cotton bolls. Although the combination of multiple-view geometry methods and ground mobile platforms allows cost-effective acquisition of high-precision 3D structures at the scale of plant organs, ground platforms still have limitations, including low efficiency, soil compaction damage, and reliance on manual operations (Sun et al., 2020).

In recent years, the swift progress of Unmanned Aerial Vehicles (UAVs) has facilitated high-throughput acquisition of plot-scale plant traits such as plant height (Che et al., 2020) and canopy coverage (Cheng et al., 2022). Unlike ground platforms, UAVs offer high image capture efficiency, negate the need for manual intervention, and prevent soil compaction. The UAV nadir route, a technique in which the lens is 90° vertically down to acquire images, is the most commonly used UAV technique for acquiring plot-scale traits (Shu et al., 2021; Shu et al., 2022). The method has also been used for the detection of some simple plant organs, such as panicles, which are often located at the top of the stem for easy identification. As an illustration, Chang et al. (2021) utilized a UAV to gather RGB point cloud data of sorghum panicles from a height of 10 m, successfully estimating the length (R^2 = 0.61) and width (R^2 = 0.83) of panicles compared with manual measurements. Nonetheless, this UAVs flight route fails to accurately portray complex plant organs as it excludes organ data from the mid and lower sections due to its top-down perspective. Some researchers have attempted cotton boll counting and yield estimation via this method (Jung et al., 2018; Yeom et al., 2018), but have faced data loss concerning mid and lower cotton bolls due to obstruction (Dube et al., 2020).

Acquiring complete 3D point clouds of cotton bolls from the field in a high-throughput and cost-effective manner, along with accurate detection of the number, volume, distribution, and yield estimation, remains a significant challenge. On the one hand, ground platforms offer high accuracy but lack efficiency and flexibility, while UAVs nadir route provide efficiency and flexibility but fall short in capturing complete 3D point clouds (Dube et al., 2020). Hence, the unresolved challenge lies in developing a UAVs platform that integrates high precision, high throughput, and flexibility. On the other hand, current cotton boll recognition algorithms primarily rely on white cotton fiber bolls, green leaves, and brown stems as reference. For example, Dube et al. (2020) utilized depth cameras and RGB cameras to obtain RGB 3D point clouds of cotton and used color filters to identify white cotton fiber bolls, exploiting their distinct color differences from green leaves and brown stems. However, these algorithms overlook the variations in cotton varieties found in breeding materials, such as brown cotton bolls or leaves in shades of purple and red. Moreover, cotton bolls, with their pliable structures and considerable variations in shape, are more likely to cause occlusion when compared to rigid fruit structures such as mangoes. Consequently, an urgent need arises for a more robust algorithm to cater to cotton breeding materials and enable accurate segmentation of cotton bolls from complex environments.

Cross-circling oblique (CCO) route is an innovative method of oblique photography for UAVs, involving multiple single circular flight routes (Xiao et al., 2023). Currently, this technology finds primary applications in urban aerial photography and 3D reconstruction, enabling comprehensive and precise 3D structural reconstruction of the target object by capturing information from a 360° perspective. CCO routes employ nonlinear movement, providing considerable efficiency advantages over traditional five-dimensional oblique routes (Xiao et al., 2023). In the traditional five-dimensional oblique route, the UAV moves linearly and necessitates a substantial image overlap between the front and back, leading to an increased image count. Implementing the CCO route enables the UAV platform to attain high throughput, precision, and flexibility, thereby opening novel avenues for reconstructing and characterizing 3D cotton bolls in the field.

The purpose of this study was to obtain high-quality 3D point clouds of various cotton breeding materials in the field using a UAV platform that offers high precision, efficiency, and cost-effectiveness. Additionally, the study aimed to develop a complete process for characterizing cotton boll traits, including the number of bolls, volume, distribution, and yield. The specific objectives were as follows: (1) to compare the accuracy of estimating the number and volume of cotton bolls using both the CCO route and the nadir route on a UAV platform; (2) to develop a comprehensive process for characterizing diverse breeding cotton bolls, encompassing white and brown fiber bolls, as well as green, purple, yellow leaves, and branches; (3) to analyze the distribution of various cotton breeding materials using the complete cotton boll point cloud reconstructed through CCO route; (4) to predict the yield of all cotton breeding materials in the field using the optimal yield estimation model extracted from CCO-derived traits.

2. Materials and methods

2.1. Study area and experimental setup

A selection of 320 representative upland cotton germplasm materials were planted at the Cotton Experimental Base of the Biotechnology Research Institute of the Chinese Academy of Agricultural Sciences, situated in Yazhou District, Sanya City, Hainan Province ($109^{\circ}12'6.030''$,

$18^{\circ}23'10.338''$). Eighty-six of these varieties were selected for this experiment and replicated once for a total of 172 plots (Fig. 1). The planting period extended from 2021 to 2022. Seed sowing was carried out around November 1, 2021, and the harvest was conducted in April 2022. The area, Sanya City, is characterized by a tropical marine monsoon climate, with an average temperature range of $22\text{--}30^{\circ}\text{C}$. A total rainfall of 456.5 mm was recorded in 2021, while the rainfall was measured with 289.5 mm in 2022. The field experiment comprised subplots measuring 3 m in length and 0.6 m in width, encompassing an area of 2.4 m^2 each. Each subplot was planted with a single row of 10 cotton plants. The sowing of seeds was achieved through manual film covering and spot sowing. Cotton was cut from the top in March 2022 to avoid overcrowding and to keep the height of the cotton at about 1 m. Defolianters were also used to facilitate cotton harvest. All other field management procedures at the experimental site, including watering, fertilizing, weeding, and pest control, were executed in accordance with local conventional cotton cultivation methods to ensure the normal growth of the cotton materials.

2.2. Data collection and processing

Data collection was carried out on April 3, 2022. A total of 86 cotton varieties were selected, with two replicates for each variety, resulting in a total of 172 plots. The selected cotton included white and brown

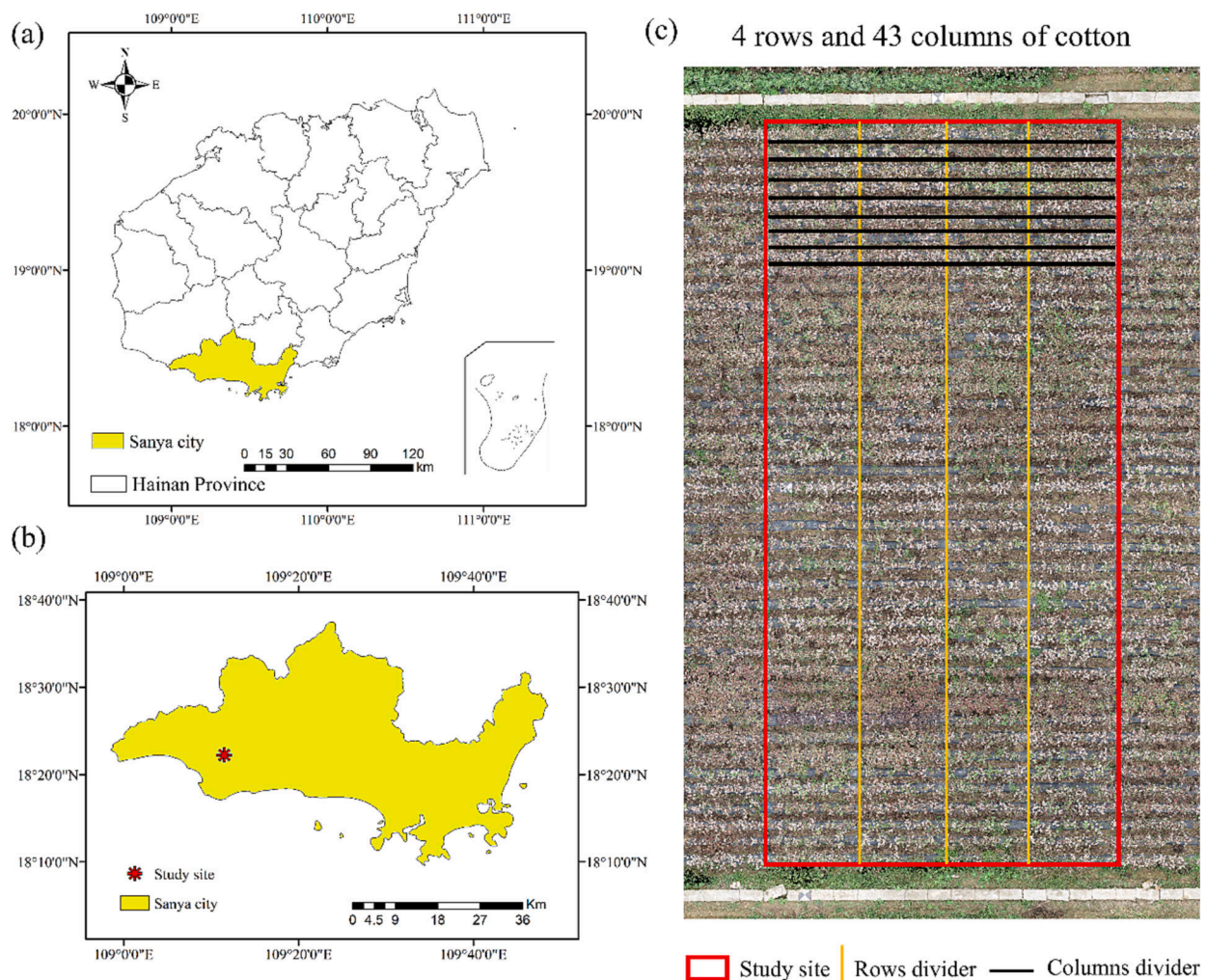


Fig. 1. (a) Indicates the geographical position of the Sanya City within Hainan Province; (b) Illustrates the specific location of the study area within Sanya City; (c) Presents the primary image of the study area, captured on April 3, 2022. The red area symbolizes the chosen 172 cotton plots, organized in a grid of 4 rows and 43 columns. The orange lines denote the separation between rows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

cotton fiber bolls, as well as green and purple leaves.

UAV images of the cotton plots were acquired using two different flight paths: a cross-circling oblique (CCO) route (Fig. 2b) and a nadir route (Fig. 2c), based on the Phantom 4 RTK (DJI, Shenzhen, China). The overlap rate within each circle of the CCO route was set to 30 photos, while the overlap rate between circles was set to 50 %. The camera angle was fixed at 45°. Regarding the nadir route, the overlap rate along the flight route was set at 80 %, while the overlap rate between routes was set at 70 %. The flight altitude for both routes was maintained at five meters above the cotton canopy. The ground sample distances (GSD) were 0.14 cm/px and 0.19 cm/px for nadir and CCO photos, respectively. Overall, there were 321 photos for the nadir route and 1063 photos for the CCO route, respectively.

Following the UAV data collection, 34 plots were randomly chosen for manually quantifying the number of cotton bolls and the weight of seed cotton. These 34 plots were evenly distributed among the 176 plots and there were no duplicates among them. Subsequently, the cotton seeds were removed to determine the weight of the lint cotton. The relationship between the measured number of cotton bolls and seed cotton and lint cotton were shown in Figure S1.

2.3. UAV data pre-processing

2.3.1. Reconstruction of 3D cotton point cloud

The software chosen for reconstructing 3D point clouds of cotton based on UAV images was Agisoft Metashape professional software (Agisoft LLC, St. Petersburg, Russia; version 1.7.0). The software utilizes the standard Structure from Motion-Multi-View Stereo (SfM-MVS) process to reconstruct 3D point clouds from multiple-view images. Structure from Motion (SfM) estimates camera positions to create a sparse point cloud, and Multi-View Stereo (MVS) refines this into a dense 3D model by analyzing images from different viewpoints (Schonberger and Frahm, 2016; Iglhaut et al., 2019). For the reconstruction of dense and precise point clouds, high-quality settings were utilized for the both photo alignment and dense point cloud generation, following the specific recommendations for vegetation (Liu et al., 2021). Statistical filtering

was employed for noise removal. In order to speed up the calculation, the voxel-based downsampling method was used to downsample the cotton point cloud, and the voxel size was set to 0.006 m.

2.3.2. Coordinate correction of 3D cotton point cloud

To facilitate further analysis, the y-axis of the 3D point cloud of cotton was adjusted to align with the long side of the plot. Initially, the 3D point cloud of cotton (Fig. 3a) was rotated 360° to obtain the point cloud representing the soil surface (Fig. 3b). Subsequently, two reference points were chosen to establish the sowing line. The angle α (Fig. 3c) formed between the sowing line and the y-axis was calculated. Equation (1) was employed to rotate the point cloud around the z-axis by an angle of α , thereby aligning the long side of the plot parallel to the y-axis (Sun et al., 2020).

$$R_z(\alpha) = \begin{pmatrix} \cos(\alpha) & \sin(\alpha) & 0 & 0 \\ -\sin(\alpha) & \cos(\alpha) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (1)$$

2.3.3. Plot segmentation of 3D cotton point cloud

Two steps were implemented for plot segmentation: row and column segmentation of the cotton point cloud. The dense planting of cotton poses challenges in determining plot boundaries from the front view. Thus, in row segmentation, the 3D cotton boll point cloud was flipped to the backside, allowing for a clearer selection of coordinates for four boundary points (Fig. 4). The cotton boll point cloud was divided into four rows based on the Y coordinates of the four boundary points. Given that cotton is cultivated on terraces, each row of 3D cotton bolls was initially divided into $N + 1$ equal sections based on their x-axis range where N represents the total number of plots in a row. The point with the lowest Z value was identified in each section, deemed the lowest point (red point in Fig. 5a). Upon dividing each row of 3D cotton bolls into plots based on the lowest point, competition between varieties emerged, resulting in certain varieties encroaching upon the living space of neighboring varieties and leading to unsatisfactory segmentation

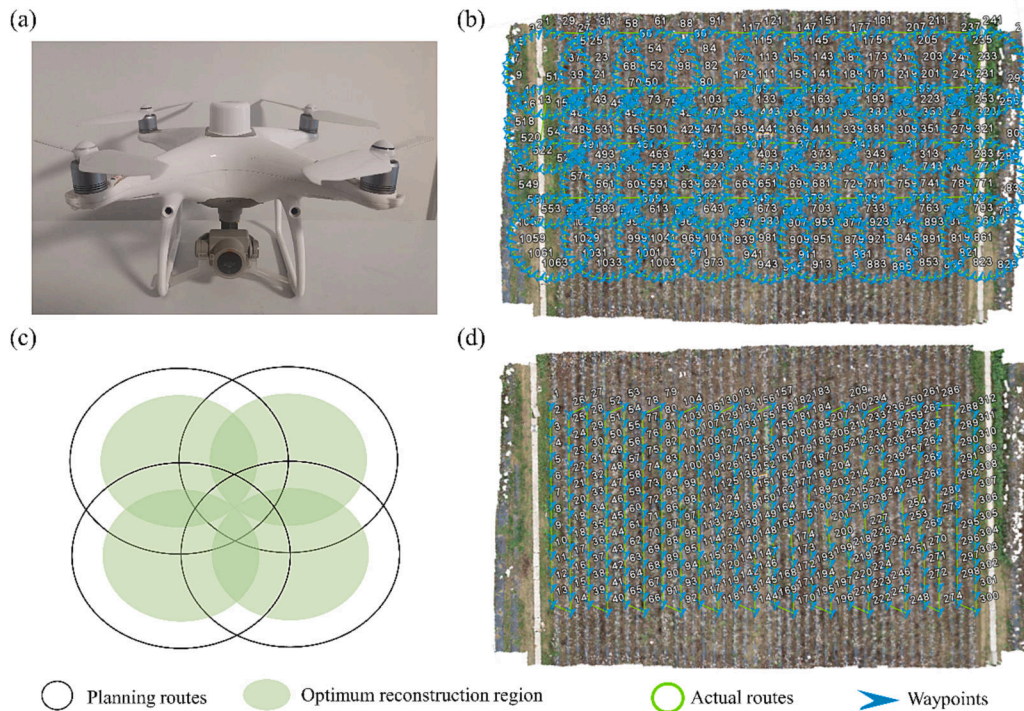


Fig. 2. (a) Schematic diagram of Phantom 4 RTK; (b) Waypoint trajectory diagram of CCO route; (c) Schematic diagram of a CCO route consisting of four single circles with a 50% inter-circle overlap; (d) Waypoints trajectory diagram of nadir route;

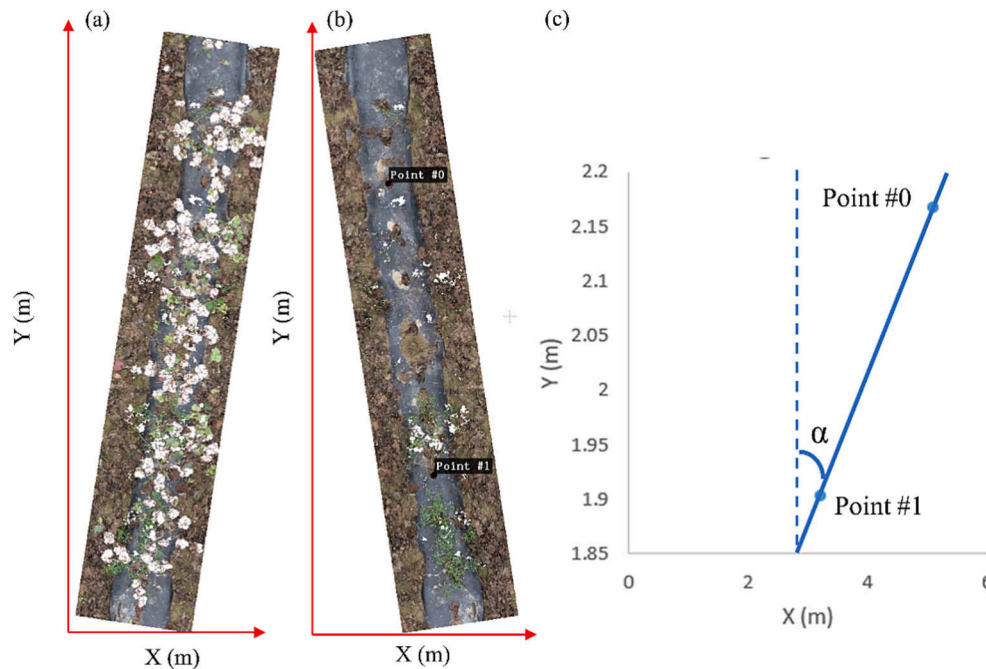


Fig. 3. (a) The front of a cotton point cloud of a plot before coordinate correction; (b) The back side of the cotton point cloud of a plot before the coordinate correction; (c) Diagram of the coordinate correction.

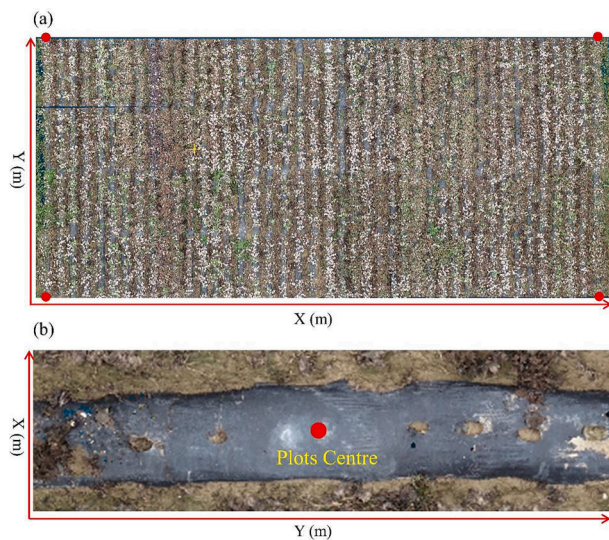


Fig. 4. Row segmentation of cotton point cloud. (a) The four selected boundary points at the front side of cotton point cloud; (b) An example of the plot centre selected from the back side of the cotton point cloud as a boundary point.

outcomes (Fig. 5b). Consequently, manual adjustment was performed on certain lowest points prior to conduction of plot segmentation, aiming to achieve the optimal column segmentation effect (Fig. 5).

2.3.4. Soil removal of 3D cotton point cloud

Upon acquiring the 3D point cloud of the cotton plot, the plant point cloud was vertically partitioned into 100 equal segments to generate its vertical distribution profile (Fig. 6). A Gaussian fitting curve with a term value of four was employed. The initial local minimum point, descending from the top, was adopted as the threshold for soil segmentation, ensuring comprehensive segmentation of the soil within the plot.

2.4. Cotton boll detection and characterization

Upon acquiring clean 3D point clouds for each individual cotton plot, an automated procedure was implemented to ascertain the count, location, and volume of cotton bolls within each plot. This process comprised two primary steps. Firstly, features pertaining to both color and morphological structure were extracted to facilitate supervised classification, distinguishing the point clouds associated with cotton bolls from those associated with branches and leaves. Subsequently, these 3D point clouds of cotton bolls were subjected to further clustering to accurately determine the quantity, position, and volume of the cotton bolls.

2.4.1. Cotton boll segmentation

(1) Super-voxel clustering of cotton point cloud

Super-voxels are a higher-level representation of 3D data that groups points with similar properties to facilitate tasks such as segmentation, object recognition, and feature extraction. Super-voxels calculation on 3D point clouds of cotton provides advantages, including improved feature extraction through voxel grouping with similar properties and preservation of object boundaries. This leads to more efficient differentiation between cotton bolls and leaf/branch point clouds, while also reducing calculation complexity. Two commonly employed super-voxel clustering methods are the Voxel Cloud Connectivity Segmentation (VCCS) algorithm (Papon et al., 2013) and the Color-based Region Growth Segmentation (CRGS) (Zhan et al., 2009). The CRGS method was selected due to the VCCS's tendency to cluster single cotton boll point clouds into numerous super-voxels during testing, which hinders the representation of the cotton boll's structural and morphological features. Conversely, the CRGS method successfully preserves the structural and morphological features of single cotton boll after the clustering process by using the merging algorithm for over- and under-segmentation control.

CRGS is a super-voxel clustering algorithm based on colorimetric similarity and spatial proximity. In this algorithm, the point cloud is first partitioned into multiple clusters based on spatial distance and color

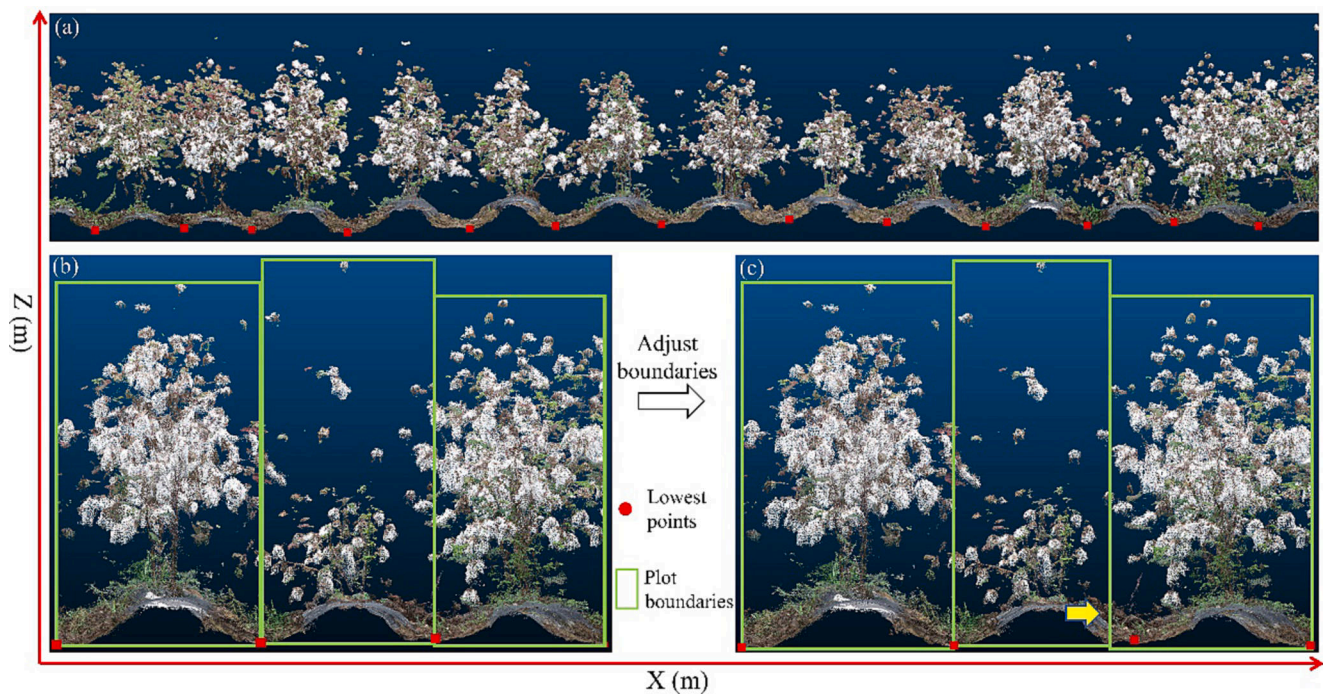


Fig. 5. Column segmentation of cotton point cloud. (a) A number of lowest points obtained from a row of cotton point clouds; (b) Results of segmenting cotton plots according to the original lowest points; (c) Results of segmenting cotton plots according to manually adjusted lowest points.

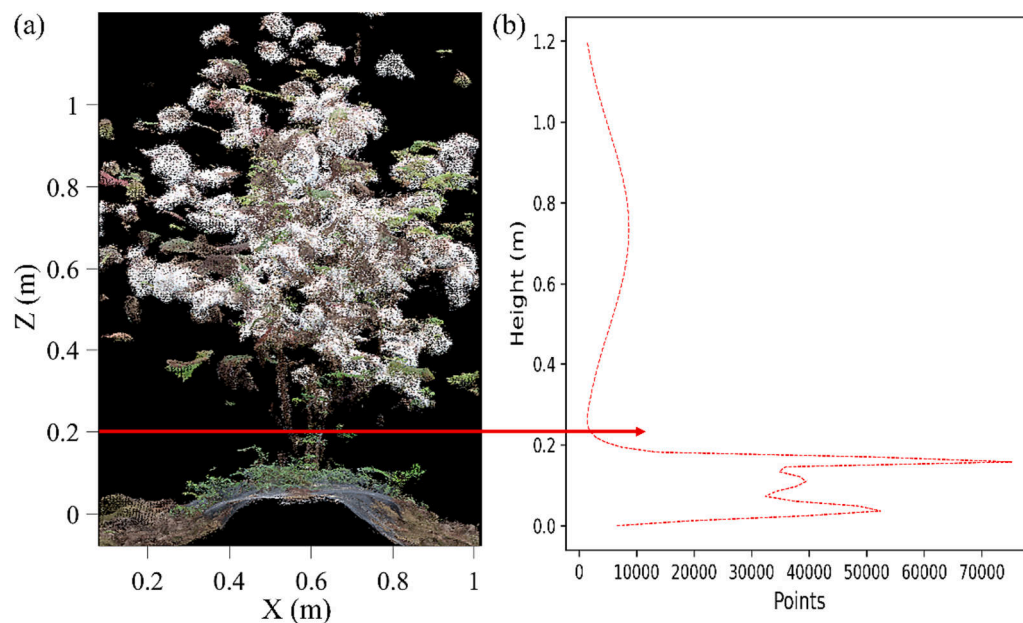


Fig. 6. Soil removal of 3D cotton point cloud; (a) An example of a cotton point cloud; (b) Vertical profile of cotton point cloud after Gaussian fitting. The position of the red arrow indicates the first local minimum position of the curve from top to bottom, which is the threshold of the soil split. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

similarity, then the clusters with similar colors are merged, and finally the clusters with too small number of points are merged. The process of CRGS is as follows: The clustering process begins by growing the first point of the input point cloud. When encountering an unlabeled point, the process initiates a new region, adds the current unlabeled point to the region, and stores it as a seed point. Subsequently, as each point is retrieved from the seed point and its k nearest neighbors are identified within the specified distance TD , the current region expands considering the colorimetric similarity threshold $C1$. Points with colorimetric simi-

larity less than $C1$ are appended to the current region and included in the seed point. The equation for computing colorimetric similarity is shown in Eq. (2). The current growth process concludes when the seed point becomes empty. The procedure generates a new region by reading each unlabeled point and repeats until all points are labeled. After the clustering process, an attempt is made to merge clusters with similar colors. Two adjacent clusters with an average color difference smaller than the threshold $C2$ are merged together. Then, the second merging step is performed. In this step, each cluster is validated based on the number of

points it contains. If this number is smaller than threshold C3, the current cluster will be merged with the nearest neighboring cluster. After testing, TD, C1, C2 and C3 were set as 0.1 m, 6, 5, 80 respectively.

$$CS(C_1, C_2) = \sqrt{(R_1 - R_2)^2 + (G_1 - G_2)^2 + (B_1 - B_2)^2} \quad (2)$$

where CS represents colorimetric similarity, C1, C2 represent color vectors of two objects which can be points and clusters. RGB values of points can be read directly from raw data while RGB values of clusters are average RGB values of the points belonging to the cluster.

(2) Feature extraction

To distinguish among clusters of cotton bolls, cotton branches, and cotton leaves, a 126-dimensional feature vector was extracted from the clusters. This vector includes color, shape, and spatial characteristics. In this study, cotton of different varieties exhibited different colors in their bolls and leaves, making the color features indispensable. A 102-dimensional color feature, extracted from RGB and HSV spaces, was presented. Here, the R, G, B, H, S, V channels were each represented by 16-bin histograms, supplemented by a 6-dimensional vector composed of the mean values of R, G, B, H, S, V. Studies suggested that HSV space can better resist the interference of light. Shape and spatial features were also essential, given the potential confusion caused by the color resemblance between brown cotton bolls and branches, as well as the white appearance that some branches and leaves may adopt under sunlight, possibly leading to misidentification with white cotton fiber bolls.

Cotton bolls exhibited significant shape differences compared to branches and leaves, thus the three eigenvalues ($\lambda_0 > \lambda_1 > \lambda_2$) from the covariance matrix of the clusters are chosen as shape features (Sun et al., 2020). These three eigenvalues denoted the variance in the three principal component directions of the clusters. Given that cotton bolls clusters approximated a circular shape across three directions, the three eigenvalues of this cluster should be similar ($\lambda_0 \approx \lambda_1 \approx \lambda_2$). As the shape of branches and leaves is linear, their three eigenvalues would exhibit notable differences. In branches, the length significantly exceeds both width and height, resulting in a pronounced disparity among the eigenvalues: $\lambda_0 \gg \lambda_1 \approx \lambda_2$. For leaves, the length and width greatly surpass the thickness, leading to eigenvalues: $\lambda_0 \approx \lambda_1 \gg \lambda_2$.

The global radius-based surface descriptor (GRSD) (Warchol 2018) was chosen to describe the spatial characteristics of the cluster. The radius-based surface descriptor (RSD) encodes the radial relationship of the point and its neighborhood. The point has the largest RSD value when it is located on a flat surface and the smallest RSD value when it is located on a curved surface. GRSD is a 21-dimensional histogram calculated by computing RSD for all points of the cluster, which is used to portray the spatial characteristics of the cluster.

(3) Deep-forest based classifier

A deep-forest classifier (Zhou and Feng, 2018) was used for binary classification of cotton bolls and branches/leaves. The deep forest classifier is an alternative to deep learning, which stacks multiple layers of random forests in a cascading manner to achieve better feature representation and learning performance. Compared to deep learning, its advantages include fewer required sample sizes, almost no need to adjust hyperparameters, and the ability to determine model complexity in a data-related manner without the need for backpropagation. The method divides the training into two stages: Multi-Grained Scanning and Cascade Forest. Multi-Grained Scanning generates features, and the prediction results are obtained through multiple layers of Cascade Forest. Specifically, in the first stage, a sliding window is used to divide into multi-instance feature vectors, which are output as class probability vectors through forest transformation. Then merge the class probability vectors to generate new features. In the second stage, the input features

are output as a class probability vector through the forest, and connected to the original input as the next layer output. After multiple cascades of forests, the final class probability vector is output. Finally, the average class probability vector of multiple forest outputs is calculated, and the largest class probability is taken as the prediction result. The algorithm flow chart is shown in Fig. 7.

In this study, four representative varieties of cotton were selected to train the deep-forest model, including white and brown cotton fiber bolls, as well as green, purple, red, and yellow leaves and branches. After super-voxel clustering, 598 clusters of cotton bolls were combined with 1,738 other clusters (leaves and branches) to form the training set for the deep forest model. Ten-fold cross-validation was used to assess the model accuracy on the training set. Subsequently, three different varieties of cotton were selected as the test set to evaluate the model's generalization ability. The test sets included test plot 1 with white cotton fiber bolls, yellow, and green leaves. test plot 2 with white cotton fiber bolls, and purple leaves; and test plot 3 with brown cotton bolls, yellow, and green leaves.

2.4.2. Cotton boll clustering and characterization

Following the segregation of the cotton boll point cloud from the rest (branches and leaves), further segmentation was performed to isolate individual cotton boll 3D point clouds, thereby facilitating their characterization. A preliminary segmentation of the cotton boll point cloud was achieved using clustering methodologies, followed by an additional division of the adhering cotton bolls. Subsequently, a comprehensive characterization of the cotton bolls, including their respective number, position, and volume, was taken in each plot.

The 3D density-based spatial clustering of applications with noise (DBSCAN) algorithm was used to cluster cotton boll point clouds into clusters. The algorithm classified the point clouds larger than density D in the boll point cloud into a cluster. Because the un-adhering cotton bolls have gaps between them, its density will be significantly smaller than that of the cotton boll. There are two parameters in the algorithm to determine the density D, which are the neighborhood radius R and the minimum number of points Minpts. Density D means that the number of points in the neighborhood radius R is greater than the minimum number of points Minpts. After testing, the radius R and Minpts were set to 1.5 cm and 10 in this study, respectively.

The cotton boll point cloud, post-clustering, has been divided into individual cotton bolls and adhering cotton bolls. In order to identify the adhering cotton bolls, the convex hull volumes of all cotton bolls were calculated and sorted in ascending order. Statistical filtering was used to remove outliers for each individual boll as well as adherent bolls to ensure the accuracy of volume calculations. It was observed that the sorted convex hull volumes of the cotton boll initially increased slowly, and there is a sudden increase at a certain position, which is the location of the two adhering cotton bolls. There are multiple positions where the volume suddenly increases sometimes, which could be due to multiple adhering cotton bolls. In order to determine the position of the first sudden increase, the difference sequence F and the average difference d of the cotton boll's convex hull volume were calculated. By testing, the first position in F equal to or greater than 5*d was found to be the first burst point, i.e., the adherent cotton boll.

After the identification of adhered cotton bolls, the quantity within each cluster was determined. For each plot, the average of the top 50 % of individual cotton boll volumes was utilized as the representative average volume. This is predicated on two primary considerations: the potential obstruction of cotton bolls by leaves, leading to incomplete reconstruction and underestimation of their convex hull volumes. Utilizing the larger-volume cotton bolls for average calculation compensates for this bias. Furthermore, in clusters of adhered cotton bolls, gaps between bolls are included in the total convex hull volume calculation, resulting in an aggregate volume exceeding the sum of individual volumes. Employing the average volume of larger bolls enables a more precise estimation of the number in attached clusters. The cotton boll

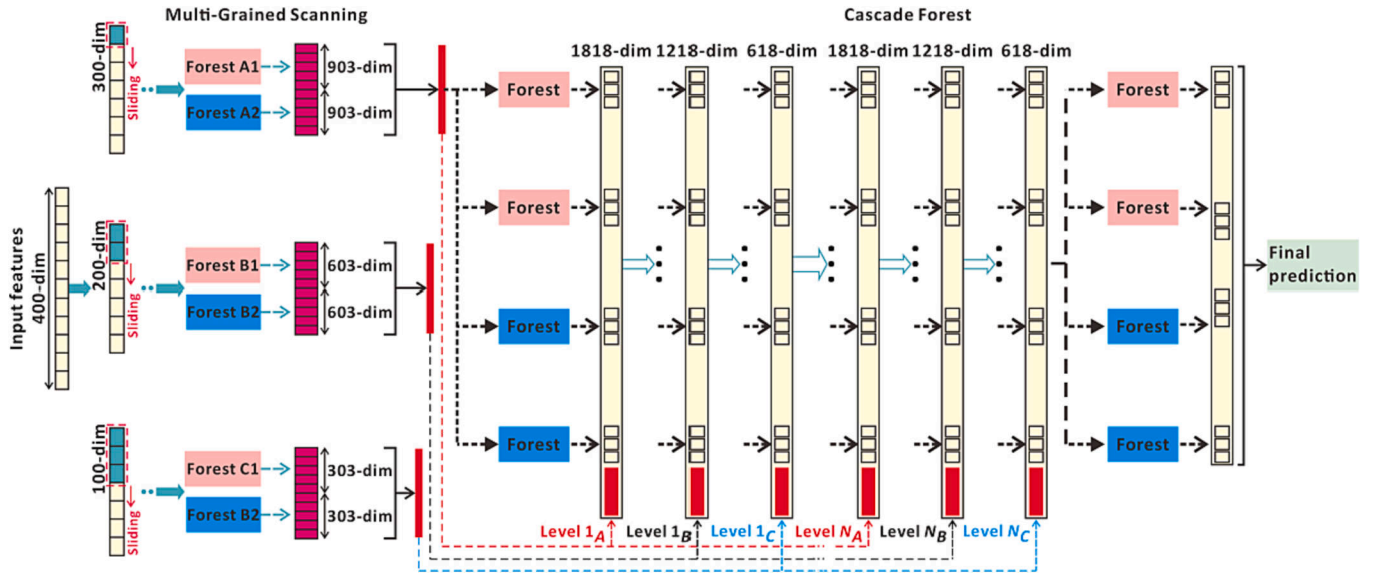


Fig. 7. Flow chart of deep-forest algorithm. The method is divided into two stages: Multi-Grained Scanning and Cascade Forest. Multi-Grained Scanning generates features and Cascade Forest cascades through multiple forests to produce prediction results.

number in adhered clusters was determined using Eq. (3). Subsequent to the quantification in each cluster, a 3D k-means algorithm segmented individual bolls. Ultimately, the convex hull volume of each boll was computed, serving as a proxy for its actual volume.

$$N = \text{round}(M/V) \quad (3)$$

where N represents the number of adhered cotton bolls, M is the convex hull volume of the adhered cotton bolls, and V is the average convex hull volume of a single cotton boll, the round function is a mathematical function used to round a number to the nearest integer.

2.5. Validation and statistical analysis

To evaluate the accuracy of our proposed algorithm, the number of cotton bolls calculated from the CCO-derived cotton point cloud and from the nadir-derived cotton point cloud were compared to the measured number of cotton bolls using the coefficient of determination (R^2), root mean square error (RMSE), and relative root mean square error (rRMSE) (Formulas 4, 5, and 6).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

$$\text{rRMSE} = \frac{\text{RMSE}}{\bar{y}_i} \quad (6)$$

where n is the number of validation samples, y_i is the manually measured value, $\hat{y}_i$ is the estimated value, and $\bar{y}_i$ is the mean value of the manually measured value.

A study on the spatial distribution of cotton boll density was carried out by utilizing the positional information of the cotton bolls within each plot. The spatial domain was partitioned into ten equal segments in the vertical (Z-axis) and horizontal (X-axis) directions, respectively. Subsequently, separate statistical analyses were conducted on the total number and volume of cotton bolls within each segment.

A linear regression modeling analysis was then conducted to explore the correlation between the total number and volume of cotton bolls within each plot and the weight of cotton seeds and lint. The best

predictive models for seed cotton and lint weight were then applied to all plots, creating maps of seed cotton and lint weight for the entire field.

3. Result

3.1. 3D reconstruction of cotton point cloud

A total of 172 cotton plots were reconstructed both derived from CCO and nadir routes. Fig. 8 depicted the CCO-derived and nadir-derived point clouds for three selected test plots. Test plot 1 included white cotton fiber bolls, yellow leaves, and green leaves. Test plot 2 included white cotton fiber bolls and purple leaves. Test plot 3 included brown cotton bolls, yellow leaves, and green leaves. The CCO-derived point clouds for cotton exhibited significantly higher density compared to the nadir-derived point clouds, thereby enabling a more accurate characterization of the morphological structure of cotton bolls. Most of the nadir-derived point clouds for cotton bolls were incomplete due to obstructions caused by branches and leaves when observed from the nadir perspective. In comparison to test plots 2 and 3, the reconstructed cotton bolls in test plot 1 exhibited higher completeness, suggesting the influence of leaf occlusion on both CCO and nadir flight routes.

3.2. Cotton boll segmentation and clustering

The cotton boll point clouds were separated from branches and leaves using Deep-forest classifier based on color, shape, and spatial structure features. Fig. 9 showed the segmentation results of the cotton bolls in three test plots. Visually, the white and brown cotton fiber bolls were effectively separated from point clouds of different colors representing branches and leaves. From a statistical perspective (Table 1), when using the deep learning classifier based on color, shape, and spatial structure features, the training set achieved a classification accuracy of 100 %, and the validation set reached 98.69 %. The model also demonstrated strong generalization ability on test set (test plots 1, 2, 3). The discrimination accuracy for white cotton fiber bolls was as high as 100 % and 99.78 %, respectively. The discrimination accuracy for brown cotton bolls also reached 95.90 %. When the Deep-forest classifier relied solely on color features, the discrimination accuracy for white cotton fiber bolls remained largely unchanged, while the discrimination accuracy for brown cotton bolls showed a slight decrease (94.63 %).

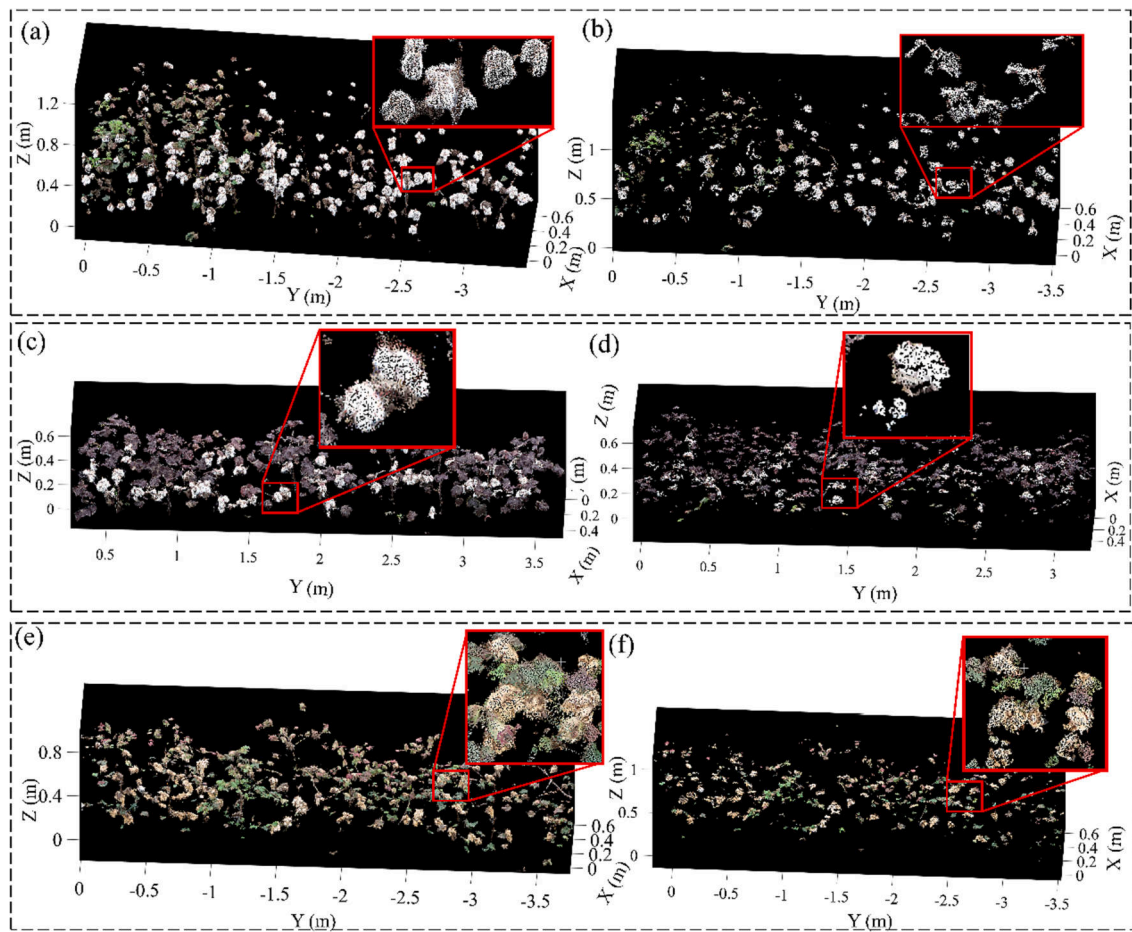


Fig. 8. CCO-derived and nadir-derived cotton point cloud for three test plots. (a) CCO-derived cotton point cloud for test plot 1; (b) nadir-derived cotton point cloud for test plot 1; (c) CCO-derived cotton point cloud for test plot 2; (d) nadir-derived cotton point cloud for test plot 2; (e) CCO-derived cotton point cloud for test plot 3; (f) nadir-derived cotton point cloud for test plot 3.

Fig. 10 displayed a scatter plot of cotton boll convex hull volumes sorted from smallest to largest, along with a line graph of their differences. Through a threshold D , the position of the adhered cotton bolls can be very effectively identified. Fig. 11 shows the cotton boll counting results for test plot 1. Each individual cotton boll was accurately segmented, which was crucial for obtaining information about the number of cotton bolls, convex hull volume, and their spatial positions. The segmentation results of adherent cotton bolls with different numbers can be find in Fig. S2.

3.3. Cotton boll counting and yield estimation

The proposed algorithm achieved good accuracy in calculating boll numbers based on both the CCO-derived and nadir-derived cotton point clouds ($R^2 > 0.73$ for both, Fig. 12). The accuracy of cotton boll counting using CCO-derived cotton point clouds was higher than that of the nadir-derived cotton point clouds (R^2 : 0.92 vs 0.73, RMSE: 7.27 vs 13.09, rRMSE: 8.57 % vs 15.40 %). The convex hull volume of the cotton bolls calculated from CCO-derived data was significantly larger than that of the nadir-derived calculation (average convex hull volume: 8729.00 cm^3 vs 3196.36 cm^3). This is because the nadir-derived cotton point clouds miss many parts of the cotton bolls during reconstruction. This indicated that the CCO flight route was more advantageous than the nadir flight route for a complete 3D characterization of cotton bolls. Combined with the proposed algorithm, it can accurately characterize cotton bolls.

The number and volume of cotton bolls calculated from CCO-derived and nadir-derived point clouds were both used to estimate the yield of seed cotton (Fig. 13) and lint cotton (Fig. 14). Overall, in estimating both

seed and lint cotton, the overall accuracy of traits calculated using the CCO-derived point cloud was significantly higher compared to those calculated using the nadir-derived point cloud. Specifically, the best estimation for seed cotton was the boll numbers calculated from the CCO-derived point cloud ($R^2 = 0.70$, rRMSE = 18.17 %). The precision of the CCO-calculated volume to estimate seed cotton was slightly lower than that of the CCO-calculated number of bolls ($R^2 = 0.67$, rRMSE = 18.97 %), but higher than the precision of the nadir-calculated trait to estimate seed cotton ($R^2 < 0.62$, rRMSE > 20.27 %). In contrast, the best for estimating lint was the boll volume calculated by CCO-derived point cloud with $R^2 = 0.75$ and rRMSE = 19.30 %. The estimation of lint cotton based on the number of cotton bolls calculated by CCO-derived point cloud had slightly lower accuracy ($R^2 = 0.65$, rRMSE = 22.57 %), but it was higher than the accuracy of estimations based on the nadir-derived point cloud ($R^2 < 0.55$, rRMSE > 25.90 %). Estimating seed cotton yield based on the number of bolls was more accurate while estimating lint based on boll volume was more accurate. This may be due to the fact that seed cotton is more correlated with the number of bolls because it contains the weight of the seed, while lint cotton does not contain the weight of the seed, so it is more correlated with the boll volume. The above results showed that the CCO-derived point cloud combined with the proposed algorithm has great advantages for the estimation of seed and lint cotton.

3.4. Bolls spatial density distribution analysis

Boll distribution, including the distribution of bolls number and boll volume, is a very important trait that is essential for variety selection.

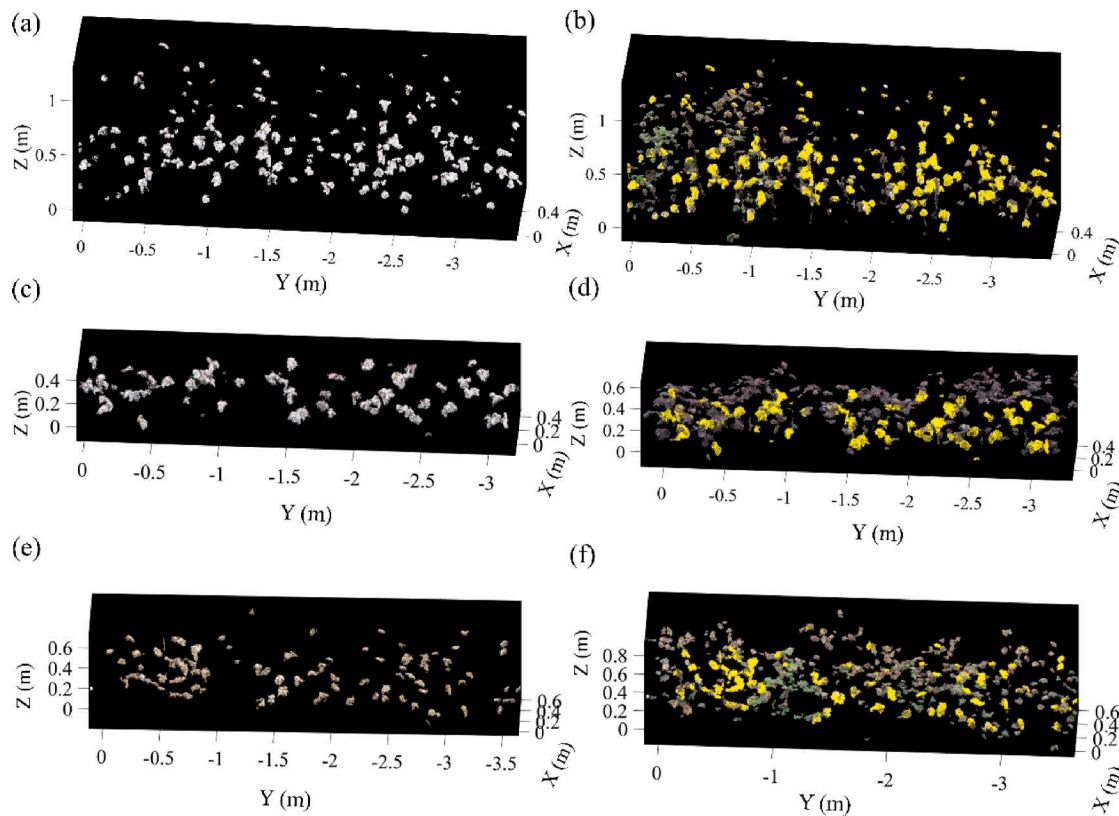


Fig. 9. Results of cotton bolls segmenting from branches and leaves in three test plots. (a) cotton bolls of test plot 1 after segmentation; (b) cotton bolls with branches and leaves of test plot 1 where the bolls were labeled yellow; (c) cotton bolls of test plot 2 after segmentation; (d) cotton bolls with branches and leaves of test plot 2 where the bolls were labeled yellow; (e) cotton bolls of test plot 3 after segmentation; (f) cotton bolls with branches and leaves of test plot 3 where the bolls were labeled yellow. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Accuracy of deep-forest classifiers based on different number of features. Note: The training data was divided into training and validation sets based on ten-fold cross-validation, and the average accuracy of ten-fold was used as the final accuracy. Three additional cotton plots (test plot 1, 2 and 3) were selected as test set to test the generalization ability of the constructed deep-forest classifier.

Deep-forest classifier	Training data		Test set		
	Training set	Validation set	Test plot 1	Test plot 2	Test plot 3
all features (126-dimension)	100 %	98.69 %	95.90 %	100 %	99.78 %
color-based feature only (102 dimension)	100 %	97.67 %	94.63 %	100 %	99.43 %

The combination of CCO routes and proposed algorithms could efficiently analyzes the distribution of cotton bolls. Upon examining the horizontal distribution, it was observed that the number and size of fiber bolls increase as they approach the main stem (Fig. 15c and Fig. 15e respectively). Particularly in the first and second segments adjacent to the main stem, the formation of cotton fiber bolls exhibited a pronounced increase (surpassing 30 % of the total), and the quality was noteworthy, contributing to over 24 % of the overall volume. However, when moving away from the main stem, the number of cotton bolls decreased and their size became smaller. This is due to the fact that the boll formation time of the bolls close to the main stem is synchronized with the most favorable climatic conditions for boll development (Gwathmey and Clement, 2010). In addition, cotton bolls close to the main stem received more organic and inorganic nutrients than those on the outside (Kannan, 2010). In terms of vertical distribution, the number

of cotton bolls was highest in the center, followed by the lower part and lowest in the upper part (Fig. 15d). The peak in the middle of Fig. 15d was shifted toward the lower part, which was due to the fact that the counts start from the ground. These horizontal and vertical distribution patterns are consistent with the statistics conducted by the expert in the field (Pettigrew, 2004; Zhang et al., 2021). However, there was no clear pattern in the vertical distribution of boll volume (Fig. 15f), which contradicts the known results (Sun et al., 2020). This may be due to the fact that late rains and winds resulted in greater deformation of the upper and lower bolls than the middle bolls, which resulted in the upper and lower boll volume being close to those of the middle layer.

3.5. Cotton yield mapping

Cotton yields were estimated for 172 plots (Fig. 16) by the best yield estimation model in section 3.3. Seed cotton was estimated by CCO-calculated bolls number while lint cotton was estimated by CCO-calculated bolls volume. The average weight of seed cotton in 172 plots was 540.36 g, while the average lint quality was 168.86 g. There was a remarkable consistency in the weight of seed cotton over lint ($R^2 = 0.93$). Cotton yields varied considerably across varieties, ranging from 127.08 g to 952.68 g for seed cotton and from 72.53 g to 291.40 g for lint cotton. The yield of cotton was mainly influenced by climate and variety.

4. Discussion

Our study demonstrated that UAV platforms are capable of mapping high-quality 3D color point clouds of cotton bolls at the field scale. Reconstructing and characterizing 3D cotton bolls present greater challenges in comparison to panicles (e.g., wheat, rice, and sorghum

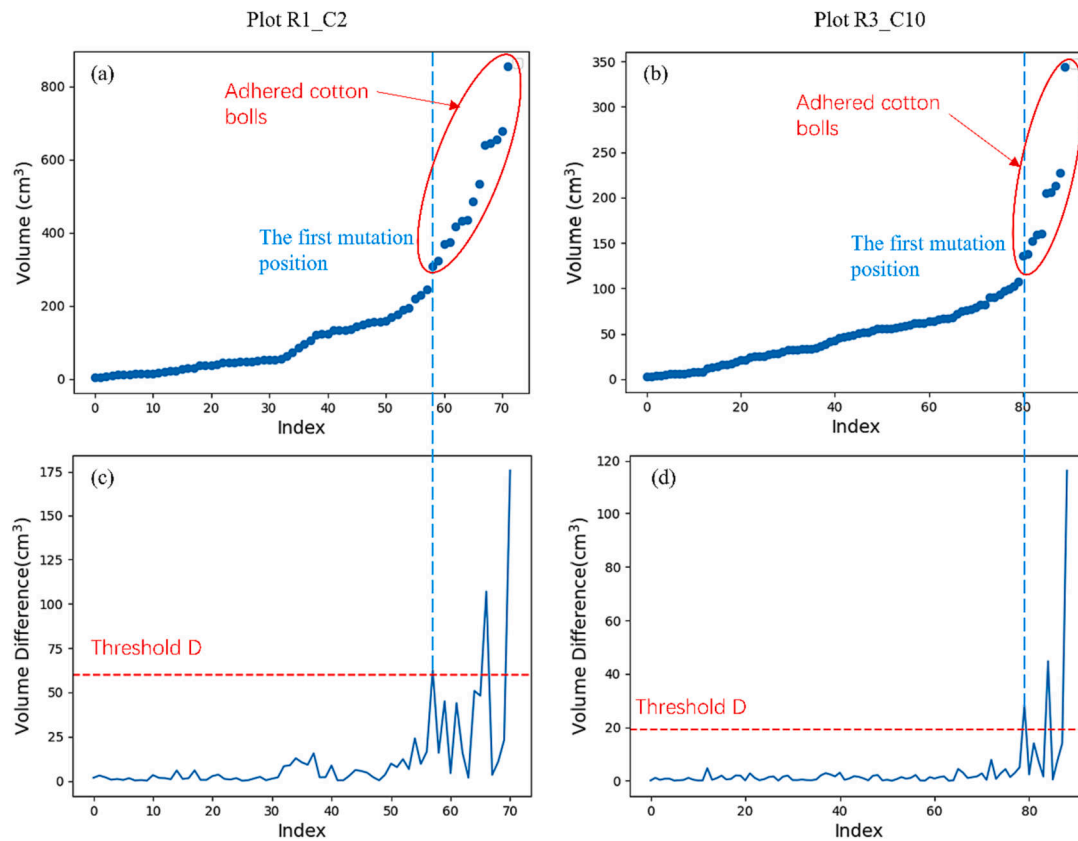


Fig. 10. Scatterplot of cotton boll convex hull volume sorted from smallest to largest and line graph of its differences. Threshold D was calculated by multiplying the average difference in cotton boll convex hull volumes by 5. It was used to determine the first mutation point, which is the position of the adhered cotton bolls. Red circles indicate adherent cotton bolls. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

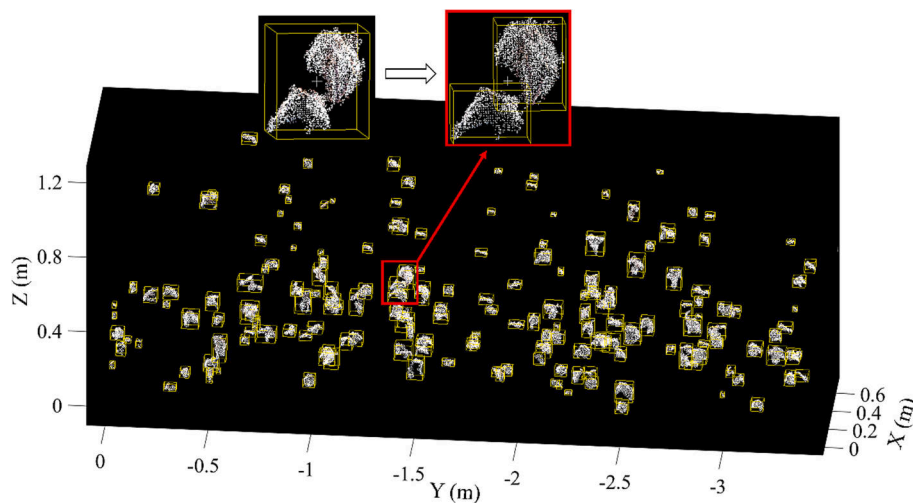


Fig. 11. Result of 3D bolls counting, each boll was indicated using a yellow bounding box. The two enlarged bounding box indicate the display from adherent bolls to individual bolls. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

panicles) due to topmost position on the stem of the latter's, which facilitates their identification. While cotton bolls exhibit a complex and overlapping distribution. Presently, the majority of studies characterizing 3D cotton bolls were conducted using ground platforms (Dube et al., 2020; Sun et al., 2020). The commonly employed methods utilize ground vehicles equipped with laser systems or multi-view camera systems. Although both methods can generate high-quality 3D point cloud of cotton bolls in the field, they suffer from lower efficiency, potential soil disturbance and compaction, and reliance on manual

operations (Maja et al., 2016; Jung et al., 2018; Yeom et al., 2018). This study utilized lightweight UAV in conjunction with a CCO flight route to achieve high-precision 3D reconstruction of the cotton bolls in the field. This approach provides several advantages, including high efficiency, low cost, flexibility, and convenience, offering a novel method for characterizing complex plant organs in 3D.

UAVs nadir flight route were conventionally employed for the 3D characterization of plant panicles, e.g., sorghum panicles, with favorable outcomes (Chang et al., 2021). This study compared the effectiveness of

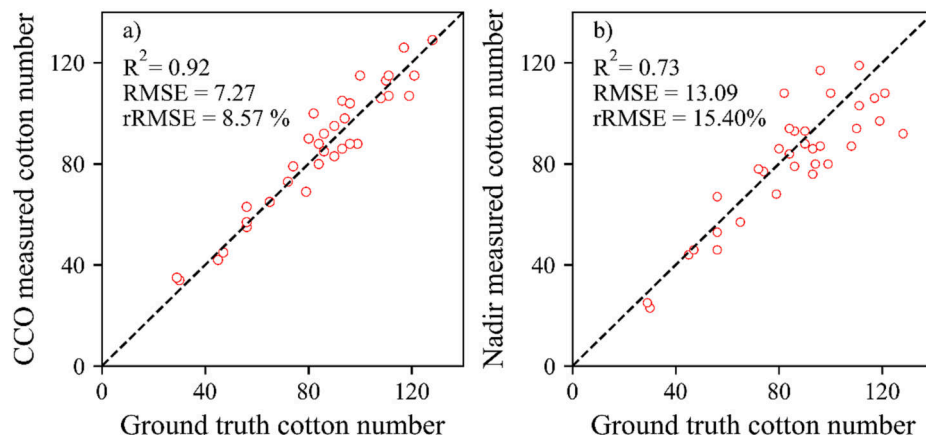


Fig. 12. Scatter plot of the calculated versus the measured boll numbers. (a) CCO-based calculated versus measured boll numbers; (b) nadir-based calculated versus measured boll numbers.

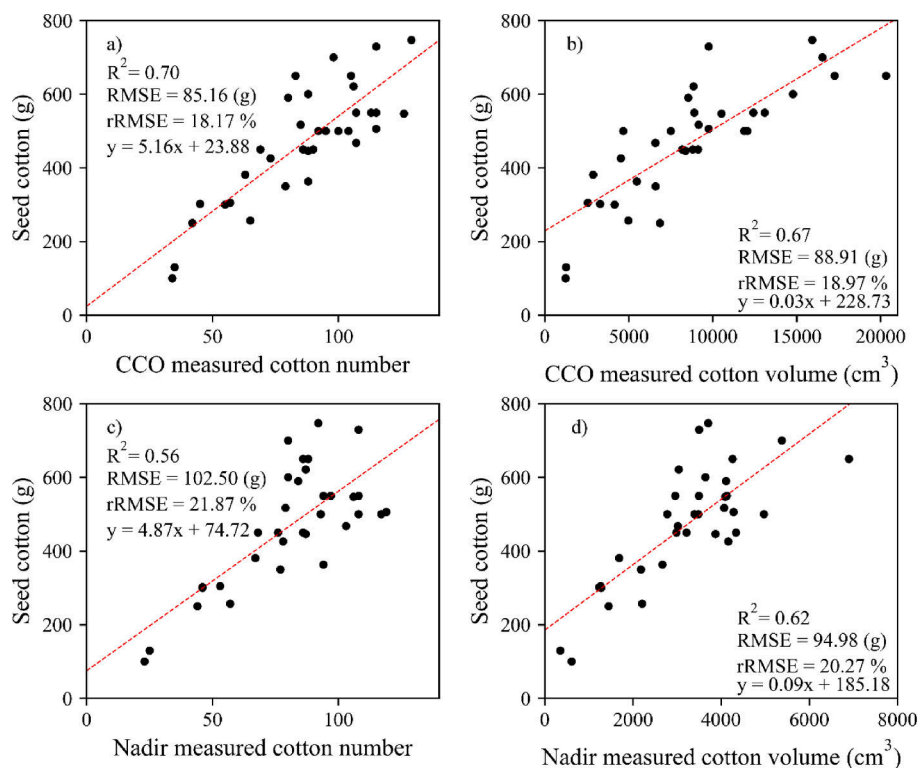


Fig. 13. The relationship between the calculated number and volume of cotton bolls, and the weight of cotton seeds. (a) CCO-based calculated boll numbers; (b) CCO-based calculated boll volume; (c) nadir-based calculated boll numbers; (d) nadir-based calculated boll volume.

the conventional nadir flight route and the CCO flight route, both positioned at the same height (5 m above the canopy), characterizing field cotton bolls in 3D. The cotton bolls reconstructed using the traditional nadir flight route exhibited fragmentation (Fig. 8b, d, f), whereas those reconstructed using the CCO flight route appeared more intact (Fig. 8a, c, d). Moreover, the accuracy of yield estimation, derived from the CCO-calculated traits, surpassed that of the conventional nadir-calculated method. This is due to the ability of the CCO flight route to capture images from all 360° angles, thereby effectively mitigating the influence of mutual occlusion among cotton bolls, a capability absent in the traditional nadir flight route. This unequivocally illustrated that the UAV-based CCO method integrates the exceptional efficiency of UAV platforms with the remarkable precision of ground platforms.

The study proposed a complete algorithmic process for characterizing 3D cotton bolls, including their quantity and volume. The

involvement of multiple cotton varieties and complex scenarios (with cotton bolls classified as white and brown, and leaves classified as green, brown, and purple) poses challenges in characterizing the cotton bolls. The deep-forest model that integrates color, morphology, and spatial features were employed to distinguish cotton bolls from branches/leaves. The results demonstrated that the deep-forest classification model, considering color, morphology, and spatial features comprehensively, achieves the highest performance on the validation set (98.69 %) and test set (>95.90 %). However, the color-based deep-forest classification model also achieves acceptable results on the validation set (97.67 %) and test set (>94.63 %). Specifically, distinguishing white cotton fiber bolls from branches/leaves relies solely on color features, while the inclusion of morphology and spatial features only improves the accuracy of distinguishing brown cotton bolls. Previous research has emphasized the importance of morphology and spatial features due to

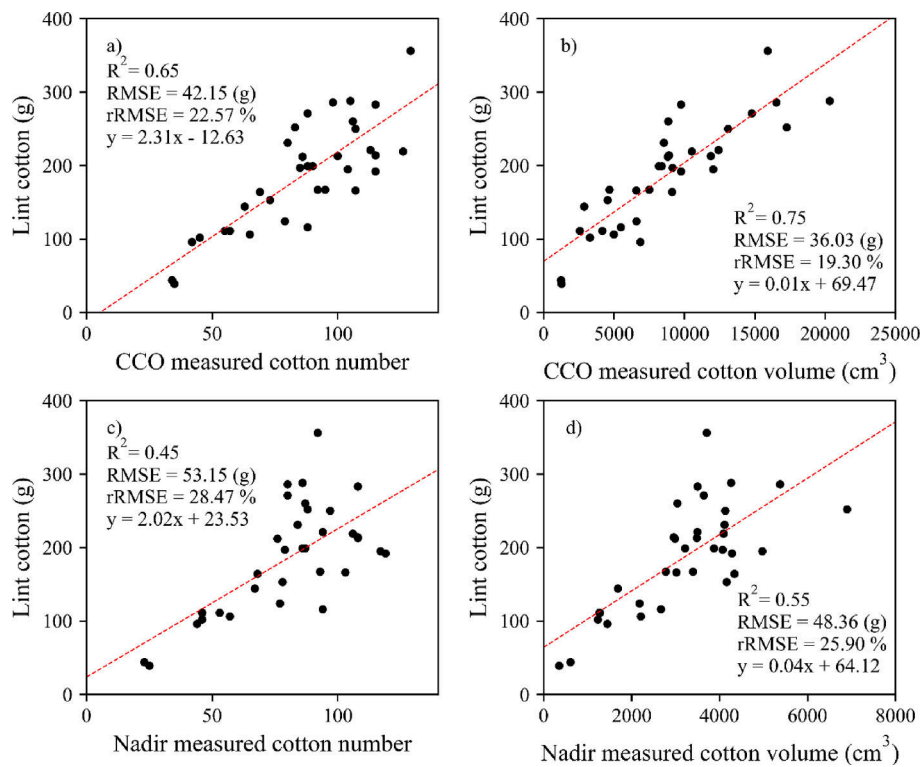


Fig. 14. The relationship between the calculated number and volume of cotton bolls, and the weight of cotton lint. (a) CCO-based calculated boll numbers; (b) CCO-based calculated boll volume; (c) nadir-based calculated boll numbers; (d) nadir-based calculated boll volume.

their ability to extract rich features unaffected by lighting conditions, although they did not compare color features separately from morphology and spatial features (Sun et al., 2020). The observed phenomenon could stem from the fact that prior research obtained multi-angle images during the midday with the sun's peak, resulting in pronounced specular reflections that rendered the branches or leaves white, thereby underlining the significance of morphological and spatial characteristics. In contrast, our study followed the experience of previous 3D reconstruction research (Liu et al., 2021; Xiao et al., 2023) and selects data collection periods with cloudy or overcast weather to avoid specular reflections, which significantly degrade the quality of 3D models. Without specular reflections, the colors of cotton bolls and branches/leaves were more easily distinguishable, benefiting the color-based classification model. Our experiment was conducted in a field trial area for cotton boll breeding, hence the density of the cotton bolls was relatively low (with a plant spacing of 30 cm). However, we are optimistic about the application of our pipeline in more densely planted cotton boll plots, as our improved method can effectively separate multiple adhered cotton bolls, as demonstrated in Figure S2. Additionally, our current cotton boll plot was partially obscured by leaves, leading to incomplete cotton boll reconstruction. We speculate that in denser plots without leaf occlusion, more complete three-dimensional cotton bolls could be reconstructed, indicating that our pipeline may exhibit superior performance. In the future, we plan to conduct more in-depth experiments in Xinjiang, China, testing the effectiveness of our pipeline in denser cotton plots.

Three-dimensional deep learning is a potential method for the next stage of exploration. The potential advantage of 3D deep learning in the characterization of 3D cotton bolls lies in its ability to highly capture spatial information and automate feature learning (Jiang et al., 2020). This means it can identify and interpret the complex three-dimensional shapes and structures of cotton bolls, providing accurate reconstruction and counting. Deep learning can also automatically learn the complex features of cotton bolls, not depending on manually designed features, greatly increasing its applicability and accuracy (Saeed et al., 2023).

However, this method also has some potential drawbacks. Firstly, training a 3D deep learning model usually requires a large amount of labeled data, which may be difficult to obtain in practice (Hinterstoisser et al., 2018). Secondly, the computational and memory requirements of 3D deep learning models are high, which may create significant demand for computational resources (Zhang and Zhu, 2018). In addition, the interpretability of 3D deep learning models is poor, which may make it difficult to understand the internal decision-making process of the model (Li et al., 2022).

The environment greatly influences the 3D characterization of cotton bolls, especially light and wind (Hu et al., 2019; Xiao et al., 2020; Xiao et al., 2021). Lighting conditions are among the environmental factors that most directly affect 3D reconstruction. The intensity, angle, and quality of sunlight can directly affect image acquisition, and thus affect the quality and precision of the reconstruction (Seifert et al., 2019). If the light is too strong or too weak, it might cause overexposure or underexposure in the images, affecting their quality (Feng et al., 2021). Changes in lighting conditions can also cause variations in the color and texture of the object's surface (Zhang et al., 2018). In this study, to avoid the influence of specular reflection on the 3D reconstruction and discrimination of cotton, images were taken on a cloudy day, which mainly results in diffused reflection. Strong wind may cause the crop to sway, preventing precise matching between consecutive images, leading to stratification of the point cloud in 3D reconstruction or even failure (Zhu et al., 2020). A relatively windless environment were chosen in this study for image acquisition, ensuring the accuracy of the model's reconstruction.

This article possesses certain limitations. Within the process of plot segmentation, a small number of neighborhoods are interconnected, and the segmentation method in this article cannot completely separate them. This, in turn, may result in errors during the subsequent cotton boll counting phase. The complete separation of these interlinked neighborhoods poses a notably formidable task. It is our anticipation that advancements in deep learning may pave the way for the resolution of this issue in future endeavors. Environmental factors, such as wind, in

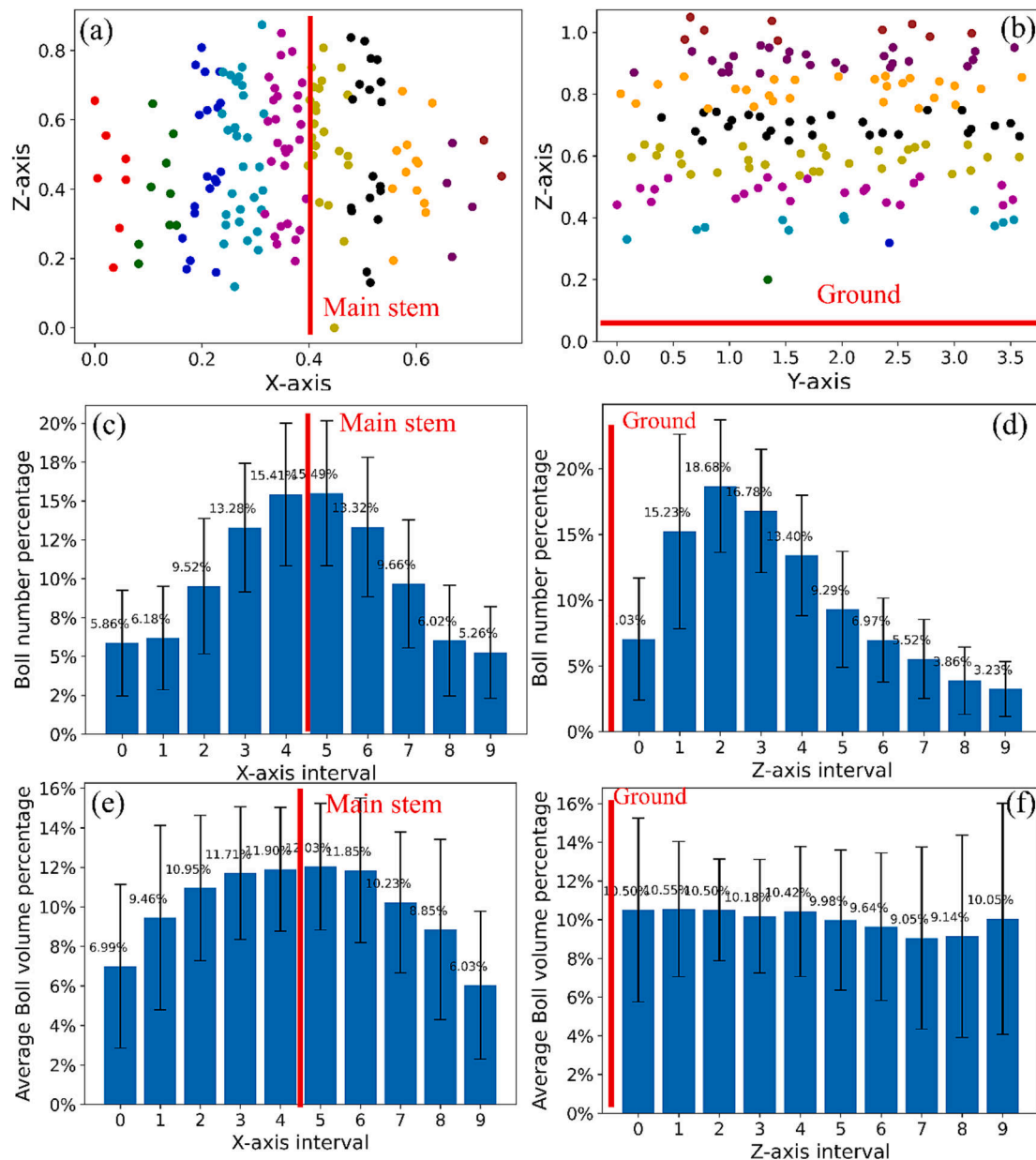


Fig. 15. Spatial density distribution of cotton bolls. Each point represents a boll (a) schematic diagram of the horizontal distribution (x-z plane) of cotton bolls; (b) schematic diagram of the vertical distribution (y-z plane) of cotton bolls; (c) Histogram of bolls number in horizontal direction (x-z plane); (d) Histogram of bolls number in vertical direction (y-z plane); (e) Histogram of bolls volume in horizontal direction (x-z plane); (f) Histogram of bolls volume in vertical direction (y-z plane).

agricultural fields, can induce morphological alterations and size increase in a minority of edge-located cotton bolls. Such changes might lead to erroneous identification of these bolls as adhered by the existing algorithm, compromising the precision of cotton boll counts. Future developments should focus on creating a more sophisticated algorithm that can mitigate these field-induced inaccuracies, enhancing the precision in quantifying the quantity, volume, and spatial distribution of cotton bolls in situ.

UAV, as one of the low-altitude remote sensing technologies, have significant application value in obtaining crop phenotypic traits on a plot-scale. The strong timeliness and operability of UAV effectively compensate for the lack of spatial and temporal resolution in the application of satellite remote sensing in precision agriculture. The introduction of the CCO method provides a new approach for UAV technology to obtain phenotypic traits at the crop organ-scale. This paper uses this method to reconstruct the 3D cotton bolls of 172 plots

and proposes an algorithm to precisely characterize the 3D distribution of cotton bolls, realizing the high-throughput estimation of seed cotton and lint cotton in the field. The method can be effectively used in the screening of cotton boll breeding varieties, greatly improving the efficiency of breeding variety selection.

5. Conclusion

UAV-based CCO routes have been employed to generate 3D point clouds of cotton bolls in the field, which are subsequently contrasted with those obtained through traditional nadir routes. Furthermore, a robust procedure has been proposed to characterize cotton boll traits. The CCO-derived 3D point cloud of cotton bolls exhibits greater completeness compared to the nadir-derived cotton bolls, accompanied by a markedly improved accuracy in the estimation of the cotton boll number and volume. The proposed procedure can identify cotton bolls

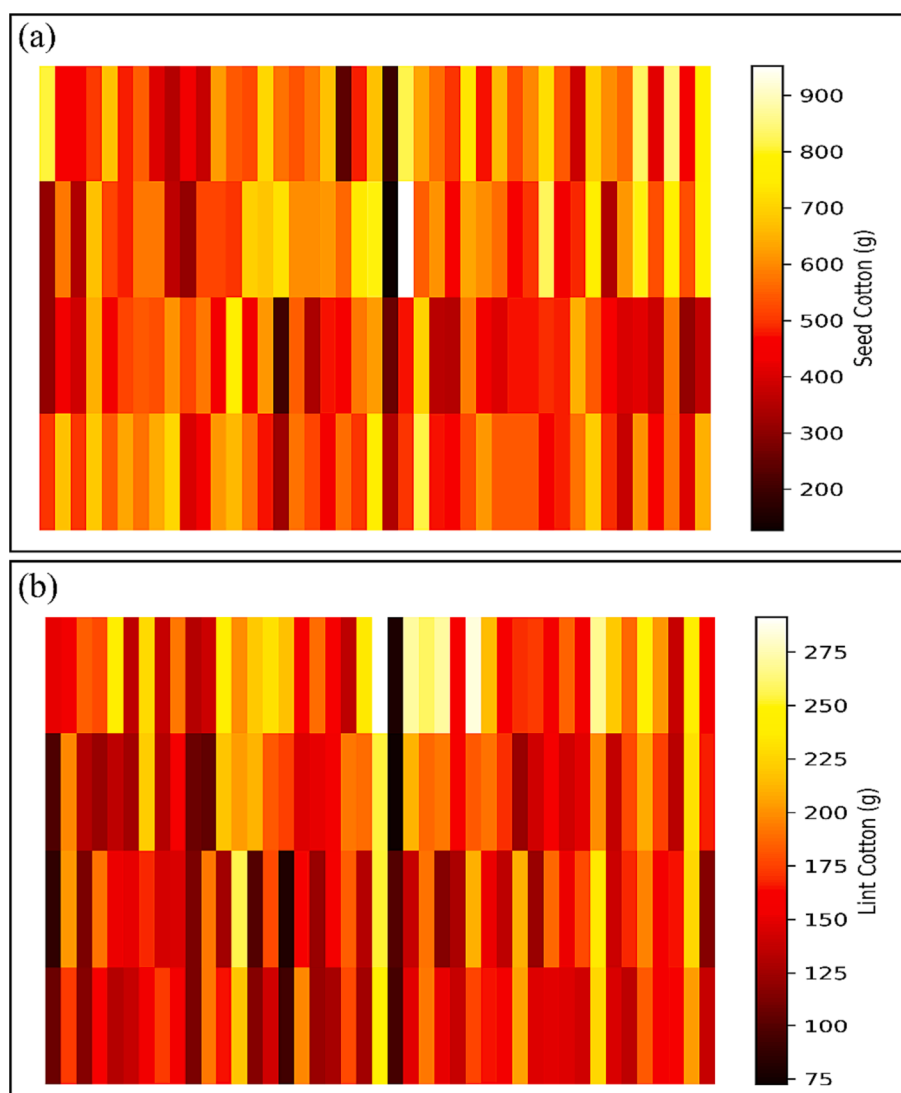


Fig. 16. The heat map of estimated cotton yield of 174 plots. (a) seed cotton mapping calculated from CCO-calculated bolls number; (b) lint cotton mapping calculated from CCO-calculated bolls volume.

of diverse colors under complex scenarios involving various cotton breeding materials with precision. Utilizing the high-precision boll point cloud regenerated by the CCO route, the in situ distribution of bolls encompassing all cotton breeding materials was determined and yield was precisely estimated. Notably, boll count exhibited superior accuracy for seed cotton yield estimation, whereas boll volume yielded higher precision in lint yield estimation. This high-throughput method delineated herein provides insights into cotton boll distribution and yield estimation, significantly aiding in the selection of superior cotton boll breeds. This study establishes the potential of the drone platform for organ-scale analysis, underscoring its role as a potent instrument for accelerating the breeding process.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.isprsjprs.2024.01.027>.

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